SmartPCB-XAI

Intelligent Optical Inspection & Analytics Platform (Industry 4.0)

INSPECTION CERTIFICATE

Generated: 2026-09-23 21:46:13

Inspection ID:	INSP-1790180135	**AI Inspection Score:**	**94.7 / 100**
Image ID:	not reported	**Pipeline Status:**	**FAIL**
Generated By:	Automated AI Inspection Engine (no human inspector)	**Score Grade:**	REJECT
Model Version:	not reported	**Defect Count:**	1

1. Detected Manufacturing Defects & Severity Ratings

#	Defect Type	Confidence	Severity Level	Severity Score	Impacted Area / Proximity
1	open	26.8%	**CRITICAL**	99.9/100	ic

2. PCB Component Verification Summary

Component Family	Detected Count	Detection Status	Note
capacitor	23	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only
diode	7	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only
ic	4	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only
resistor	59	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only
switch	1	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only
transistor	3	**DETECTED BY MODEL**	Not a placement/land-pattern verification; detection only

3. Rule-Based Root-Cause Guidance & Suggested Engineering Actions

No repair actions needed.

4. XAI Attribution — Interpretability Summary

Attribution Method	N/A — XAI payload absent from inspection_data
Scientific Status	NOT COMPUTED
Disclaimer	Run inspection with XAI enabled to populate this section.
Interpretation Guidance	N/A

5. Inference Timing & Model Metadata

Stage	Mean Time (ms, if available)
0. Preprocessing (CLAHE/resize)	N/A
1. Component YOLO Inference	N/A
2. Defect YOLO Inference	N/A
3. XAI Attribution (approx)	N/A
4. Postprocess / Severity / KB	N/A
Total Inspection Round-Trip	N/A
Model / Engine Field	**Value**
Component Model File	not reported
Defect Model File	not reported
Engine Version	not reported
Inference Device	not reported
CUDA Available (actual measured)	not reported
Component Model Class-Count Match to canonical 6?	See /health endpoint model_status.component_model.verification.class_names_match
Defect Model Class-Count Match to canonical 6?	See /health endpoint model_status.defect_model.verification.class_names_match

6. Confidence Distribution — Detection Quality

Confidence Bin	Detection Count	Share (%)
Very Low (0.00 ≤ c < 0.30)	10	10.2%
Low (0.30 ≤ c < 0.50)	28	28.6%
Medium (0.50 ≤ c < 0.70)	29	29.6%
High (0.70 ≤ c < 0.90)	27	27.6%
Very High (0.90 ≤ c ≤ 1.00)	4	4.1%
Total Detections	**98**	**100.0%**

7. Limitations & Scientific Disclaimers

7.1 XAI Approximation: The XAI overlay is an activation-energy approximation (mean absolute channel feature-map energy). It is **NOT an authentic gradient-weighted Grad-CAM++** and must not be cited as causal gradient evidence.

7.2 Severity Score is Project-Defined: All severity scores (5–100) and levels (LOW/MEDIUM/HIGH/CRITICAL) are computed using the **SmartPCB-XAI project-defined deterministic formula** documented in ai/severity/severity_engine.py. They are not an

IPC-A-610 numeric score. IPC-A-610 is used solely as qualitative workmanship guidance reference text in the repair KB.

7.3 Root-Cause = Rule-Based Recommendation Engine, Not Learned Causal Model: Probable causes and repair actions are outputs of a **Rule-Based Root-Cause Recommendation Engine** (see ai/recommendation/repair_kb.py). They are framed as *possible causes*, not proven causes. No causal inference ML model has been trained or validated.

7.4 mAP >96% is Aspirational Only: Historical claims of 97.6% component mAP or 98.2% defect mAP are **UNVERIFIED** in this audit. The only reproducible training artifacts are partial epoch CSVs (epoch-1 component mAP50=0.018, epoch-2 defect mAP50=0.456). Full training-to-convergence with a held-out leakage-free test set has not been reproduced.

7.5 Board-Level Data Leakage = PARTIALLY VERIFIED: Cross-split board-ID and filename intersection = 0 for both datasets (see DATASET_REPORT.md — NEW UPDATE 2026-09-23). Perceptual-hash (pHash) near-duplicate cross-split detection is UNAVAILABLE because the imagehash pip package is not installed; 0.0% leakage in the perceptual sense remains UNVERIFIED.

7.6 CPU-Only Benchmarked Environment: Baseline inference benchmark measured on CPU (torch 2.13.0+cpu, no CUDA). Synthetic 640×640 image mean total = 2,508 ms (N=3). Historical claim of 16.1 ms latency is UNVERIFIED — that figure would require a GPU (A100/RTX) with batch optimization, FP16, and/or TensorRT compilation, none of which have been validated in this audit environment.

7.7 Class Taxonomy: Components PARTIAL, Defects FULL/Positional (VERIFIED 2026-09-23, corrected): The deployed component checkpoint emits 10 classes (relay, buzzer, ic, switch, battery, transistor, capacitor, resistor, led, diode); the curated component dataset defines a different 6-class vocabulary — the component mapping is name-based and PARTIAL: 'inductors' exists in the curated taxonomy but no deployed class can emit it, and relay/buzzer/switch/battery/led have no curated equivalent. The deployed defect checkpoint is decoded POSITIONALLY by class index ({0:open, 1:short, 2:mousebite, 3:spur, 4:copper, 5:pin-hole}); its embedded name strings are unreliable (verified by geometric matching on val, 921/923 diagonal pairs) and it never emits 'missing_hole' — an earlier version of this disclaimer described a 'missing_hole→pin-hole' approximate alias, which was a disproven premise and is withdrawn. Defect mapping status is FULL. See DEFECT_CLASS_SCHEMA.md §7 and /health class_taxonomy.

7.8 Class Imbalance Warning: Component dataset: inductors = 620 / 33,156 total boxes (~1.87%). Defect dataset: copper = 1,474 / 10,013 (~14.7%). Minority classes may have depressed precision/recall even at full convergence. Class-balanced loss or focal loss may be warranted at retraining time.

7.9 Reports Contain No Fake Metrics: PDF / Excel reports only contain numbers actually computed from inspection_data or model safety endpoints. No placeholder performance figures (e.g. 97.6%, 16.1ms, 0.0%) are embedded into generated reports unless they have been genuinely produced by a validated run in this audit.